CHAPTER 3 ELECTROCHEMISTRY

Electrochemistry is the study of the relationship between electricity and chemical reactions.

The chemical reactions involved in electrochemistry are

REDOX REACTION:

A reaction that involves both oxidation and reduction

Reduction reaction

- Gain of electron
- Oxidation number decreases
- Example:

$$\underset{+2}{Cu^{2+}(aq)} + 2e^- \rightarrow \underset{0}{Cu(s)}$$

- Occurs at cathode

Oxidation reaction

- Loss of electron
- Oxidation number increases
- Example:

$$\underset{0}{Zn(s)} \rightarrow \underset{+2}{Zn^{2+}(aq)} + 2e^-$$

- Occurs at anode

Decrease in oxidation number

$\therefore$ **Reduction (at cathode)**

Example: $\overset{0}{Zn}(s) + \overset{+2}{Cu^{2+}}(aq) \rightarrow \overset{+2}{Zn^{2+}}(aq) + \overset{0}{Cu}(s)$

Increase in oxidation number

$\therefore$ **Oxidation (at anode)**

At Cathode

- Cu^{2+} is **reduced** to Cu
- Oxidising agent: **Cu^{2+}**

At Anode

- Zn is **oxidised** to Zn^{2+}
- Reducing agent: **Zn**

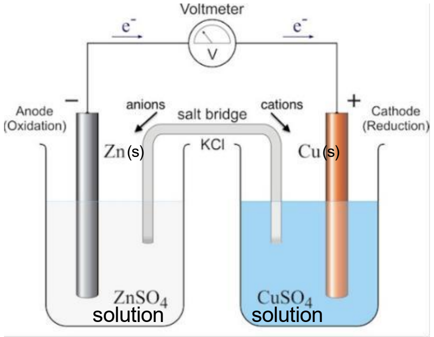
* Oxidising agent is reduced and reducing agent is oxidised.

Two types of cells in electrochemistry

Galvanic cell

- Spontaneous redox chemical reaction produces electricity.

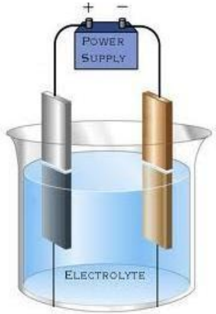
Chemical Energy → *Converted to* → Electrical Energy



Electrolytic cell

- Use electricity to cause a non-spontaneous redox chemical reaction to occur.

Electrical Energy → *Converted to* → Chemical Energy

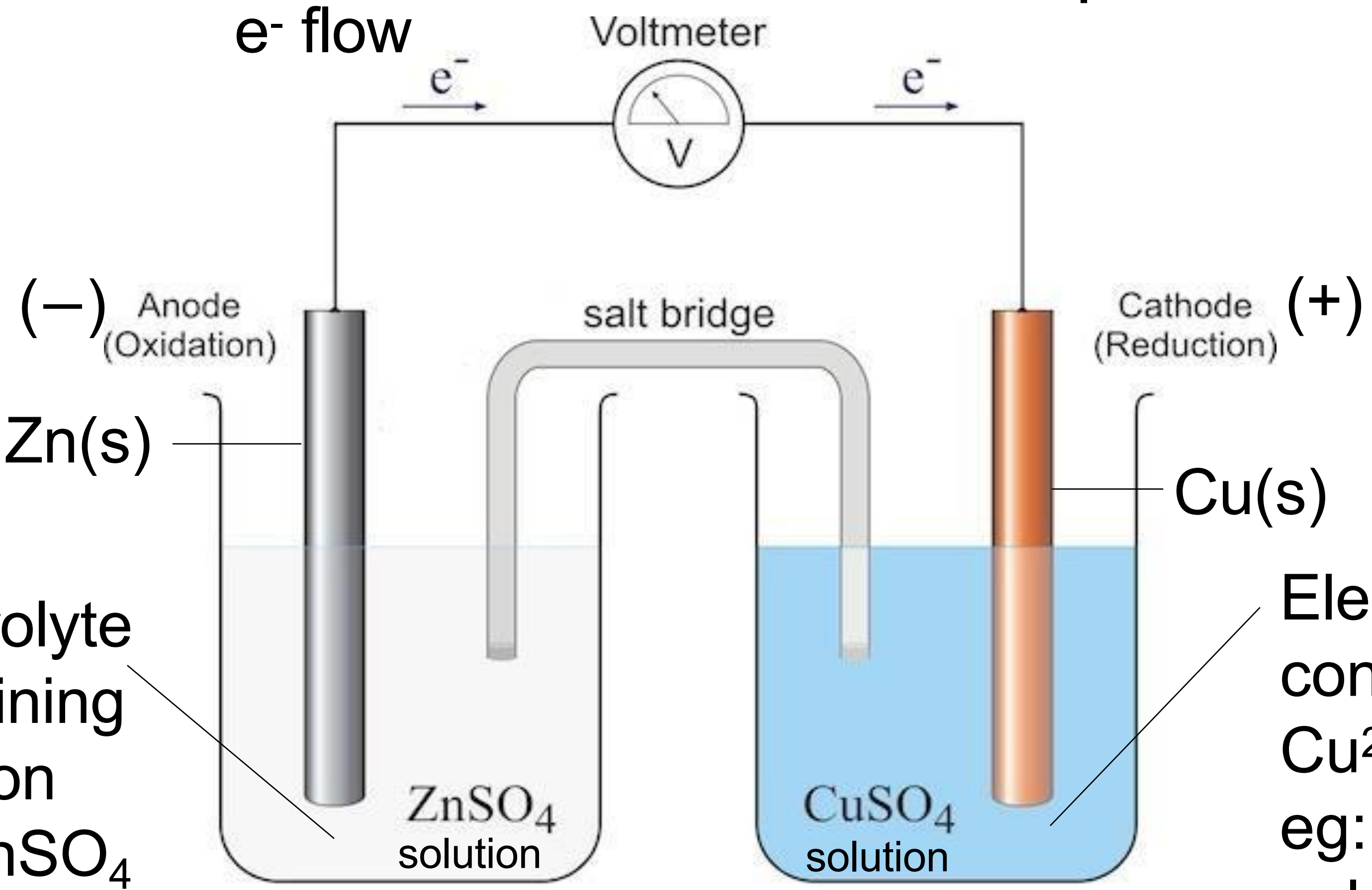


3.1
GALVANIC CELL

Galvanic Cell

- Also called voltaic cell.
- Consists of two different metals each immersed in separate beakers containing their respective metal ions in solution (electrolyte).
- Example of galvanic cell: Daniell cell which consist of:
    - Zinc half-cell (Zn metal in an aqueous solution of Zn^{2+})
    - Copper half-cell (Cu metal in an aqueous solution of Cu^{2+})
- The two half-cells are connected by a salt bridge.
- A voltmeter is used to detect voltage generated.

Example of Daniell cell

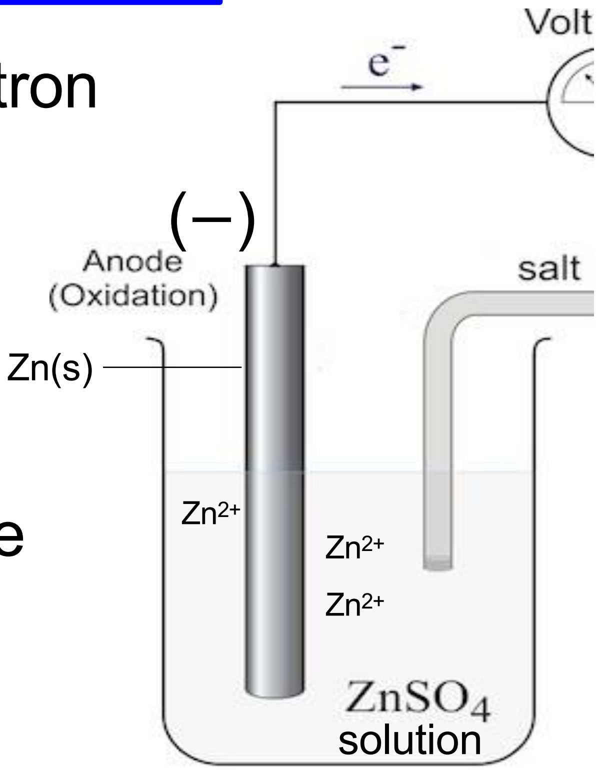


Electrolyte containing Zn^{2+} ion eg: $ZnSO_4$ solution

Electrolyte containing Cu^{2+} ion eg: $CuSO_4$ solution

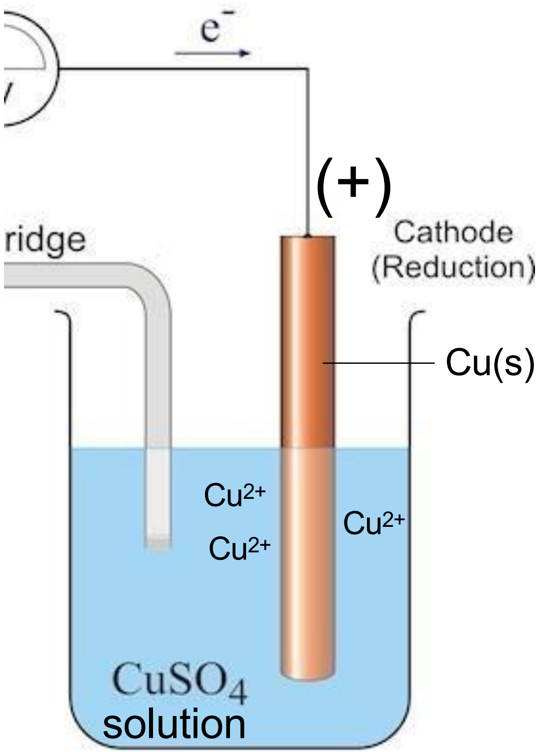
What happens at the zinc electrode?

- Zn has greater tendency to release electron (undergoes **oxidation**) than Cu.
- At anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$
  - Zn atoms lose electrons to form Zn^{2+} ions, which enter the $ZnSO_4$ solution.
  - Zinc electrode dissolves that cause the mass of zinc electrode decreases.
- Anode is **negative electrode** because it releases electrons to the external circuit.



What happens at the copper electrode?

- The electrons from the Zn metal moves through the external circuit to enter the Cu half-cell.
- Cu^{2+} ions from $CuSO_4$ solution accept the electrons (undergoes **reduction**).
- At cathode: $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$
  - Cu^{2+} ions gained electrons and reduced to Cu metal, which deposited at the Cu electrode. Mass of Cu electrode increases.
- Cathode is **positive electrode** because it gains electrons from the external circuit.



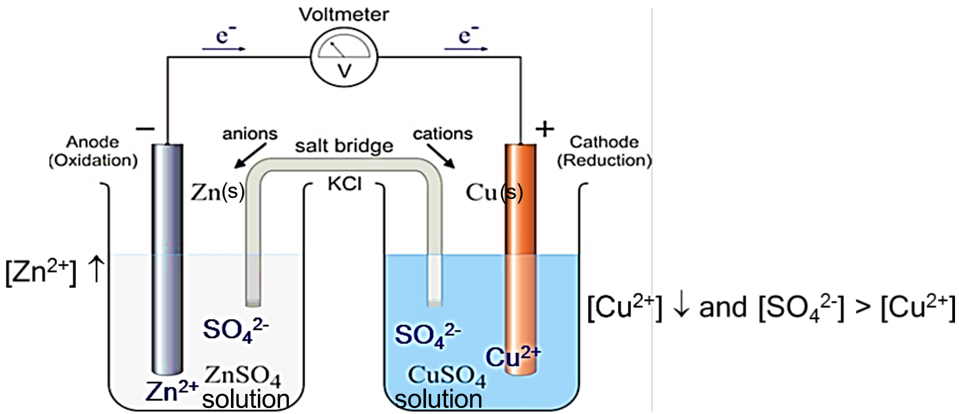
Salt bridge

- A salt bridge is an inverted U tube containing an inert salt solution such as KCl, Na_2SO_4, NH_4NO_3.
- Cations from the salt bridge move toward the cathode.
- Anions from the salt bridge move toward the anode.

Functions of salt bridge:

- to maintain electrical neutrality.
- to complete the circuit by allowing the ions to move between the two half-cells.

How does the cell maintain its electrical neutrality?



$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

- Zn^{2+} ions enter the solution, causing an overall excess of positive charge.
- **Anion**, Cl^- ions from salt bridge move into the Zn half-cell.

- Cu^{2+} ions leave the solution, causing an overall excess of negative charge.
- **Cation**, K^+ ions from the salt bridge move into the Cu half-cell.

Therefore, the electrical neutrality is maintained.

What happen if there is no salt bridge?

- If no salt bridge were present, the solution in one half-cell would accumulate negative charge (cathode) and the solution in the other half-cell would accumulate positive charge (anode) as the reaction proceeded.
- Thus, preventing production of electricity and hence the reaction will stop.
- This excess charge buildup can be reduced by adding a salt bridge.

Half-cell equation and overall equation

- The equation at cathode and anode is called the **half-cell equation.**

$$\text{Anode} \quad : \; Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

$$\text{Cathode} : \; Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

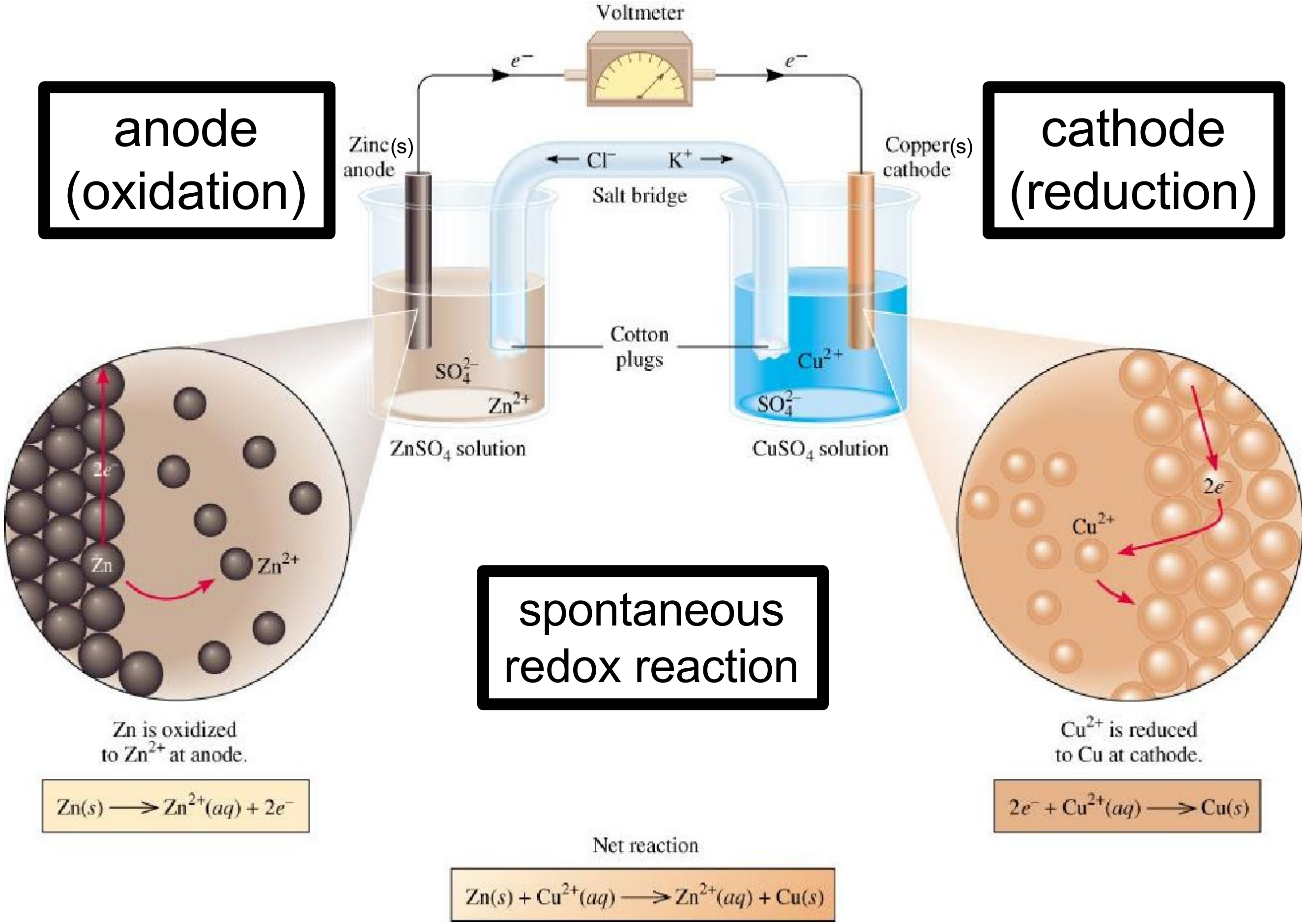
- The combination of the two half-cell equations is called an **overall equation.**

$$\text{Anode: } Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

$$\text{Cathode: } Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

Summary of a galvanic cell

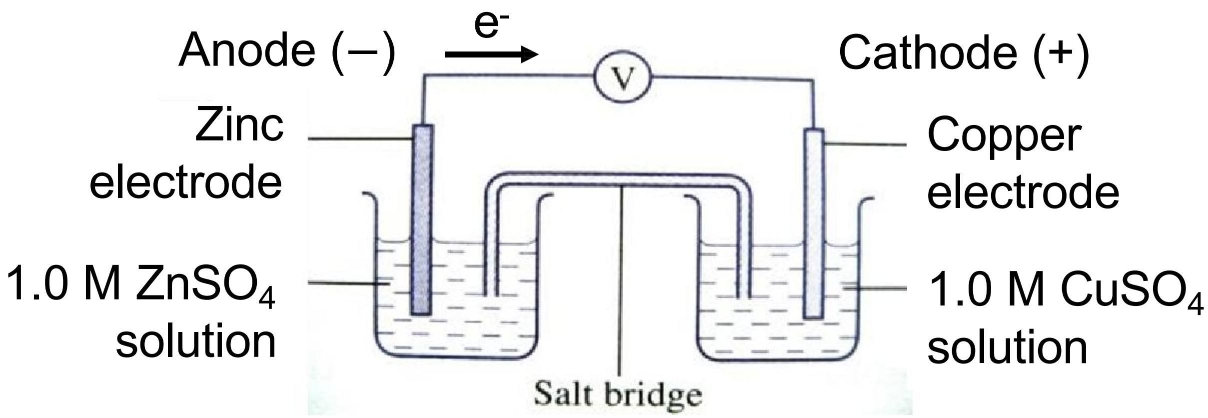


Cell Notation

- Is a shorthand method of expressing reactions in a galvanic cell.
- In a cell notation:

1. **Anode** is on the **left** and **cathode** is on the **right**.
2. The **double vertical line** represents the **salt bridge**.
3. The **single vertical line** represents the **phase boundary**.
4. Species in the same phase are separated by commas.
5. Phase and concentration are included in the cell notation.

Example:



Salt bridge

Phase boundary

$$\underbrace{Zn(s)\ |\ Zn^{2+}(aq,\ 1.0M)}_{\text{Half-cell Anode}}\ \underbrace{||}_{\text{Bridge}}\ \underbrace{Cu^{2+}(aq,\ 1.0M)\ |\ Cu(s)}_{\text{Half-cell Cathode}}$$

The ABC rule!

Standard Reduction Potential, $E°_{red}$

- The potential generated by a reduction half-reaction at an electrode measured relative to the standard hydrogen electrode under standard conditions.

- Standard conditions:
  - Temperature of 25°C (298.15 K)
  - Partial pressure 1 atm for gases
  - Concentration of ions in the electrolyte is 1 M

Standard Reduction Potential, $E°_{red}$

- Also known as:
  - Standard Electrode Potential ($E°_{red}$)
  - Standard Half-cell Potential ($E°_{half\text{-}cell}$)
- The E^o_{red} of a half-cell may have a positive or negative value.
- The half-cell is written in **reduction** form.
- The half-cell reactions are **reversible**.
- The sign of the E^o_{red} changes when the reaction is reversed.
- Changing the stoichiometric coefficients of a half-cell reaction *does not* change the value of E^o_{red}.

Table 19.1 Standard Reduction Potentials at 25°C*

Increasing strength as oxidizing agent (upward) | Increasing strength as reducing agent (downward)

Half-Reaction	E°(V)
$F_2(g) + 2e^- \longrightarrow 2F^-(aq)$	+2.87
$O_3(g) + 2H^+(aq) + 2e^- \longrightarrow O_2(g) + H_2O$	+2.07
$Co^{3+}(aq) + e^- \longrightarrow Co^{2+}(aq)$	+1.82
$H_2O_2(aq) + 2H^+(aq) + 2e^- \longrightarrow 2H_2O$	+1.77
$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \longrightarrow PbSO_4(s) + 2H_2O$	+1.70
$Ce^{4+}(aq) + e^- \longrightarrow Ce^{3+}(aq)$	+1.61
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O$	+1.51
$Au^{3+}(aq) + 3e^- \longrightarrow Au(s)$	+1.50
$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$	+1.36
$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O$	+1.33
$MnO_2(s) + 4H^+(aq) + 2e^- \longrightarrow Mn^{2+}(aq) + 2H_2O$	+1.23
$O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O$	+1.23
$Br_2(l) + 2e^- \longrightarrow 2Br^-(aq)$	+1.07
$NO_3^-(aq) + 4H^+(aq) + 3e^- \longrightarrow NO(g) + 2H_2O$	+0.96
$2Hg^{2+}(aq) + 2e^- \longrightarrow Hg_2^{2+}(aq)$	+0.92
$Hg_2^{2+}(aq) + 2e^- \longrightarrow 2Hg(l)$	+0.85
$Ag^+(aq) + e^- \longrightarrow Ag(s)$	+0.80
$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$	+0.77
$O_2(g) + 2H^+(aq) + 2e^- \longrightarrow H_2O_2(aq)$	+0.68
$MnO_4^-(aq) + 2H_2O + 3e^- \longrightarrow MnO_2(s) + 4OH^-(aq)$	+0.59
$I_2(s) + 2e^- \longrightarrow 2I^-(aq)$	+0.53
$O_2(g) + 2H_2O + 4e^- \longrightarrow 4OH^-(aq)$	+0.40
$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$	+0.34
$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$	+0.22
$SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \longrightarrow SO_2(g) + 2H_2O$	+0.20
$Cu^{2+}(aq) + e^- \longrightarrow Cu^+(aq)$	+0.15
$Sn^{4+}(aq) + 2e^- \longrightarrow Sn^{2+}(aq)$	+0.13
$2H^+(aq) + 2e^- \longrightarrow H_2(g)$	0.00
$Pb^{2+}(aq) + 2e^- \longrightarrow Pb(s)$	−0.13
$Sn^{2+}(aq) + 2e^- \longrightarrow Sn(s)$	−0.14
$Ni^{2+}(aq) + 2e^- \longrightarrow Ni(s)$	−0.25
$Co^{2+}(aq) + 2e^- \longrightarrow Co(s)$	−0.28
$PbSO_4(s) + 2e^- \longrightarrow Pb(s) + SO_4^{2-}(aq)$	−0.31
$Cd^{2+}(aq) + 2e^- \longrightarrow Cd(s)$	−0.40
$Fe^{2+}(aq) + 2e^- \longrightarrow Fe(s)$	−0.44
$Cr^{3+}(aq) + 3e^- \longrightarrow Cr(s)$	−0.74
$Zn^{2+}(aq) + 2e^- \longrightarrow Zn(s)$	−0.76
$2H_2O + 2e^- \longrightarrow H_2(g) + 2OH^-(aq)$	−0.83
$Mn^{2+}(aq) + 2e^- \longrightarrow Mn(s)$	−1.18

Standard Reduction Potential, $E°_{red}$

- The more positive the E^{o}_{red} of a half-cell, the stronger the attraction for electrons and the greater the tendency for reduction.
- Unit: volt (V).

Example:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s) \qquad E^o = +0.34\ V$$

$$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s) \qquad E^o = -0.76\ V$$

- The standard reduction/electrode potential, $\mathbf{E^o_{red}}$ for **Cu** is **more positive** than Zn.
- Therefore, the Cu half-cell becomes the **cathode (reduction)**, the Zn half-cell becomes the anode (oxidation).
- Hence,

Half-cell equation at cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Half-cell equation at anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Overall equation: $Cu^{2+}(aq) + Zn(s) \rightarrow Cu(s) + Zn^{2+}(aq)$

Standard Cell Potential, E°_{cell}

- The potential difference between the anode and cathode that causes the electrons to flow through the circuit under standard conditions (temperature of 25°C , ion concentration of 1 M and partial pressure of gas at 1 atm).
- Also known as cell voltage or cell electromotive force (emf).

$$\boxed{E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}}$$

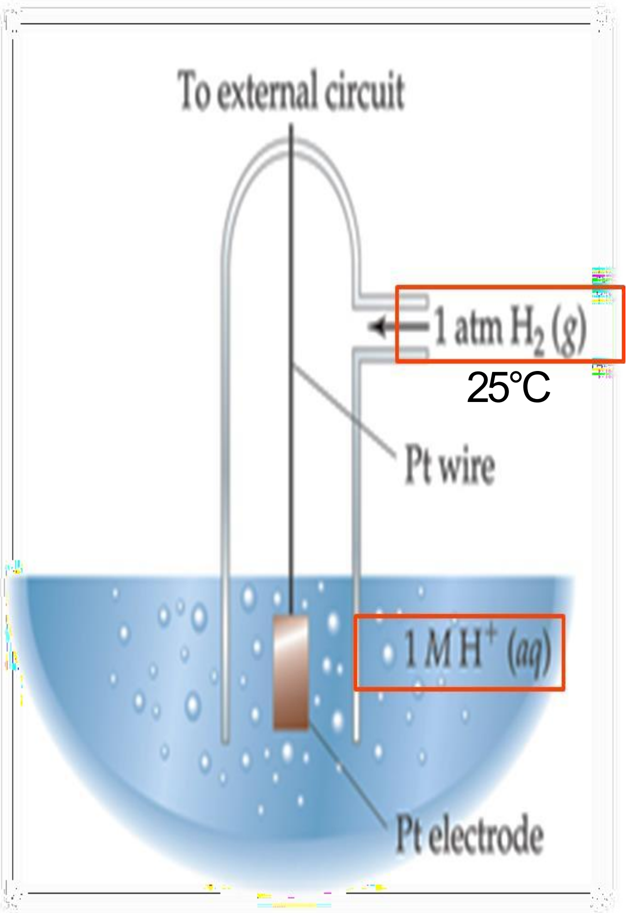
or

$$\boxed{E^\circ_{cell} = E^\circ_{red} + E^\circ_{ox}}$$

Change the sign

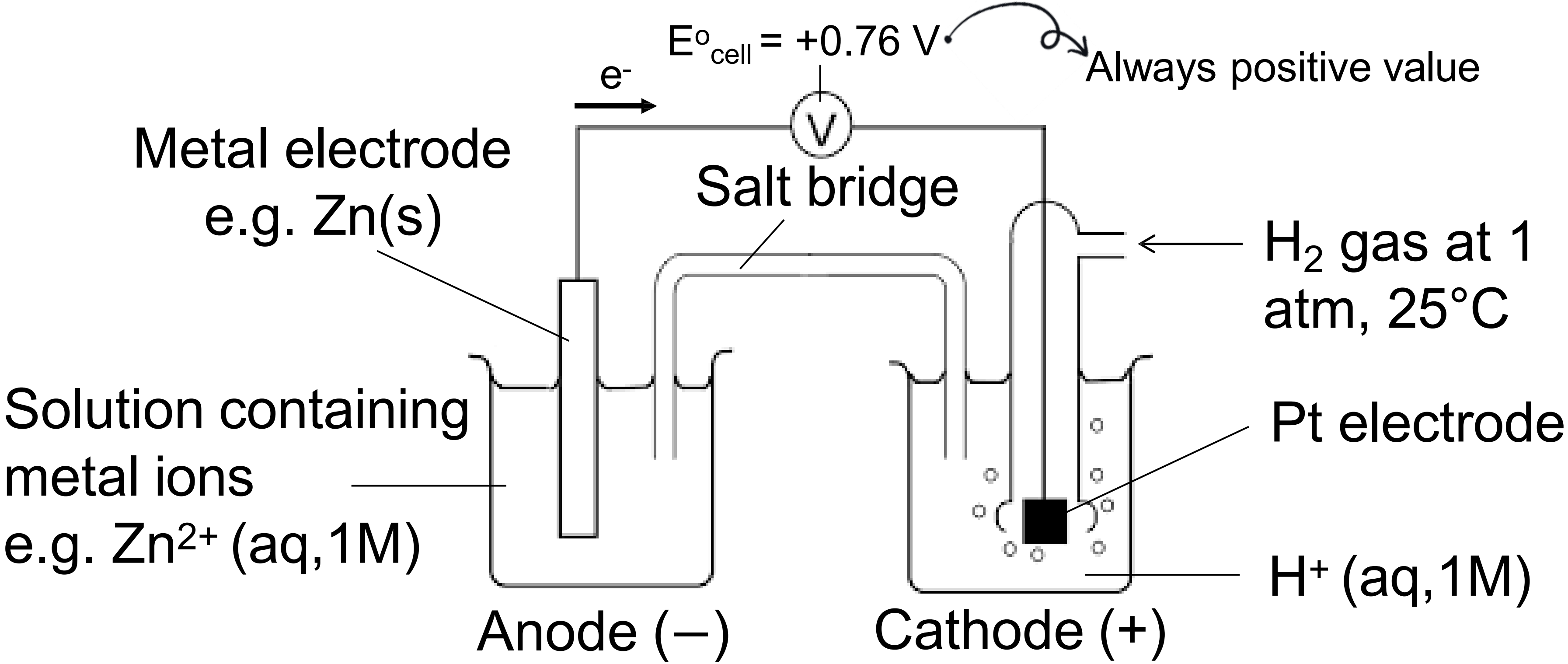
- E^o_{red} value of a half-cell is determined by measuring the electrode potential generated by a reduction half-reaction relative to a Standard Hydrogen Electrode, SHE.
- SHE is made up of a platinum electrode, immersed in an aqueous solution of H^+ (1 M) and bubbled with hydrogen gas at 1 atm pressure, and temperature at 25°C.
- The standard reduction potential of SHE = 0.00 V,

$$2H^+(aq) + 2e^- \rightarrow H_2(g) \qquad E^o = 0.00\ V$$



Using SHE to determine Standard Reduction Potential

Example: The standard reduction potential of Zn half-cell is measured by setting up the electrochemical cell as below:



Anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode: $2H^+(aq) + 2e^- \rightarrow H_2(g)$

$$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$$

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

$$E^\circ_{cell} = E^\circ_{H^+/H_2} - E^\circ_{Zn^{2+}/Zn}$$

$$+0.76\ V = 0.00\ V - E^\circ_{Zn^{2+}/Zn}$$

$$E^\circ_{Zn^{2+}/Zn} = -0.76\ V$$

* The direction of the half-cell reaction of SHE depends on the other half-cell connected to it.

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$Sn^{4+}(aq) + 2e^- \longrightarrow Sn^{2+}(aq)$	+0.13
$2H^+(aq) + 2e^- \longrightarrow H_2(g)$	0.00
$Pb^{2+}(aq) + 2e^- \longrightarrow Pb(s)$	−0.13
$Sn^{2+}(aq) + 2e^- \longrightarrow Sn(s)$	−0.14
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$\boxed{Zn^{2+}(aq) + 2e^- \longrightarrow Zn(s)}$	$\boxed{-0.76}$
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$Mn^{2+}(aq) + 2e^- \longrightarrow Mn(s)$	−1.18

Increasing strength as oxidizing agent

Increasing strength as reducing agent

- The cell notation for SHE is either:
  - $H^+(aq)\ |\ H_2(g)\ |\ Pt(s)$ when it is cathode

Eg: $Zn(s)\ |\ Zn^{2+}(aq, 1\ M)\ ||\ H^+(aq, 1M)\ |\ H_2(g, 1\ atm)\ |\ Pt(s)$

  - $Pt(s)\ |\ H_2(g)\ |\ H^+(aq)$ when it is anode

Eg: $Pt(s)\ |\ H_2(g, 1\ atm)\ |\ H^+(aq, 1\ M)\ ||\ Cu^{2+}(aq, 1\ M)\ |\ Cu(s)$

Take note that: For both cases, E^o of SHE remains zero!

The uses of Standard Electrode Potential, E^o_{red}

- To compare the relative strength of oxidising agents or reducing agents
- To calculate the value of standard cell potential, $E°_{cell}$ (to determine the anode and cathode in a galvanic cell)
- To predict the spontaneity of a redox reaction

Compare the Relative Strength of Oxidising or Reducing Agents

- The **more positive the E°$_{red}$ value**:
  - the easier the species on the left side of the reduction half-equation to undergo **reduction**.
  - the stronger the oxidising agent.
  - the half-cell will be the **cathode** in a galvanic cell.
- The **more negative the E°$_{red}$ value**:
  - the easier the species on the right side of the reduction half-equation to undergo **oxidation**.
  - the stronger the reducing agent.
  - the half-cell will be the **anode** in a galvanic cell.

Example:

Increasing strength of oxidising agent

$Cu^{2+}(aq)$	$+ 2e^- \rightarrow$	$Cu\ (s)$		$E° = +\ 0.34\ V$
$2H^+\ (aq)$	$+ 2e^- \rightarrow$	$H_2\ (g)$		$E° = 0.00\ V$
$Zn^{2+}(aq)$	$+ 2e^- \rightarrow$	$Zn\ (s)$		$E° = -\ 0.76\ V$

Increasing strength of reducing agent

Relative strength of the oxidising agent: $Zn^{2+} < H^+ < Cu^{2+}$

Relative strength of the reducing agent: $Cu < H_2 < Zn$

Exercise:

Identify the reducing agents in the half-cell equations below. Arrange the reducing agents in increasing order of strength.

$X^{3+} + 3e^- \rightarrow X \qquad E^o = -1.66\ V$

$Y^{2+} + 2e^- \rightarrow Y \qquad E^o = -2.87\ V$

$L^{2+} + 2e^- \rightarrow L \qquad E^o = +0.85\ V$

Answer:

Reducing agents: X, Y, L

Lower E^o_{red} → *easier to oxidise* → *stronger reducing agent*

Increasing order of strength of reducing agents:

L < X < Y

Calculate the standard cell potential (E^o_{cell}) of a galvanic cell

Example:

An electrochemical cell is constructed from the following two half-cell reactions:

$$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s) \quad E° = -0.76 \text{ V}$$

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s) \quad E° = +0.34 \text{ V}$$

The more positive the $E°_{red}$ value is cathode

Calculate the standard cell potential for the reaction.

Answer:

$$E^o_{cell} = E^o_{cathode} - E^o_{anode}$$

$$= E°_{Cu^{2+}/Cu} - E°_{Zn^{2+}/Zn}$$

$$= +0.34 - (-0.76) = \mathbf{+1.10\ V}$$

Predict the spontaneity of a redox reaction

$E^\circ_{cell} > 0$ (+ve)	The reaction is spontaneous
$E^\circ_{cell} < 0$ (-ve)	The reaction is not spontaneous
$E^\circ_{cell} = 0$	The reaction is at equilibrium

Example:

Given: $Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$ $E^\circ = -0.76$ V

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ $E^\circ = +0.34$ V

$2H^+(aq) + 2e^- \rightarrow H_2(g)$ $E^\circ = 0.00$ V

$Ca^{2+}(aq) + 2e^- \rightarrow Ca(s)$ $E^\circ = -2.87$ V

Example:

Determine whether the following reactions, at standard state, are spontaneous as written.

a) $Cu(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2(g)$ → Compare the changes in oxidation number to identify anode and cathode

Answer:

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

$$= E^\circ_{H^+/H_2} - E^\circ_{Cu^{2+}/Cu}$$

$$= 0.00 - (+0.34) \quad = -0.34\ V$$

Since **E°_{cell} is negative**, **the reaction is not spontaneous.**

Example:

Determine whether the following reactions, at standard state, are spontaneous as written.

b) $Ca^{2+}(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + Ca(s)$ ➔ Compare the changes in oxidation number to identify anode and cathode

Answer:

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

$$= E^\circ_{Ca^{2+}/Ca} - E^\circ_{Zn^{2+}/Zn}$$

$$= -2.87 - (-0.76) = -2.11\ V$$

Since **E°_{cell} is negative**, **the reaction is not spontaneous.**

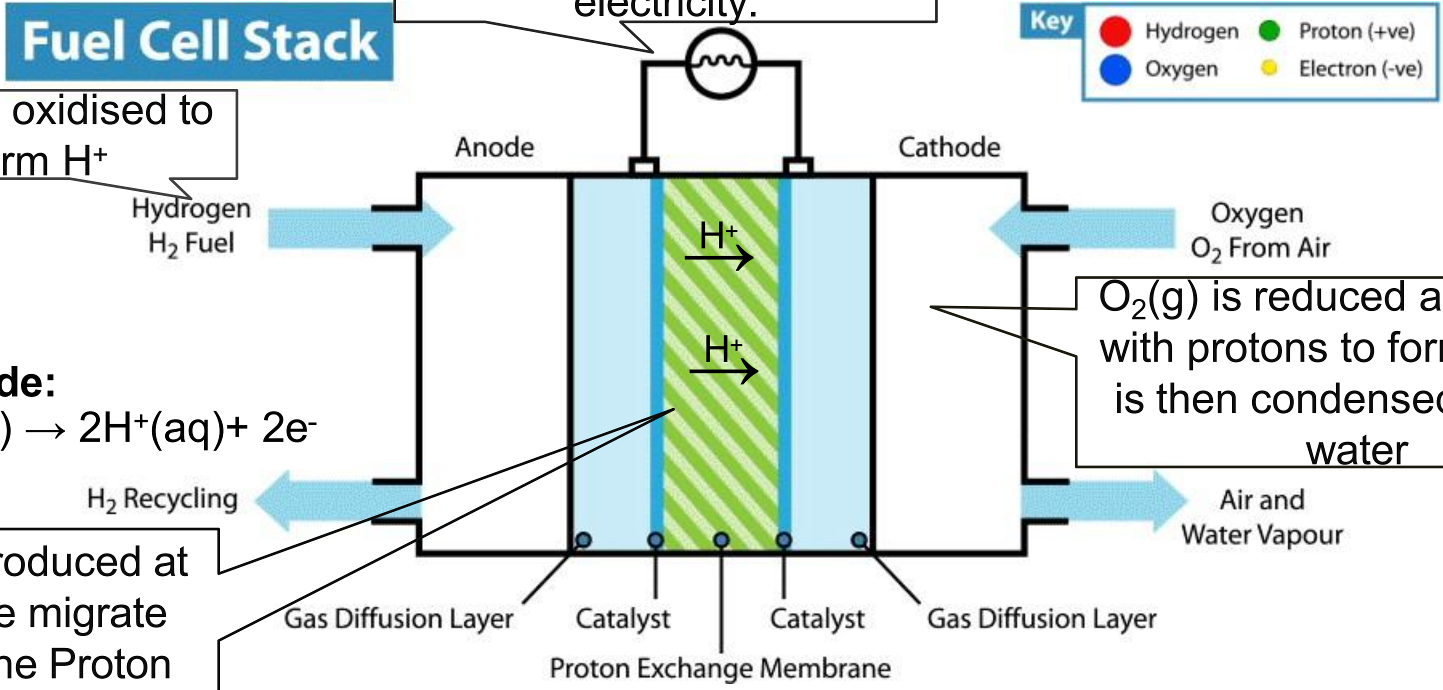
Hydrogen-Oxygen Fuel Cell

- A fuel cell uses the chemical energy of hydrogen or other fuels to cleanly and efficiently produce electricity. If hydrogen is the fuel, the only products are electricity, water, and heat.
- Fuel cells have several benefits over conventional combustion-based technologies currently used in many power plants and vehicles.
- Fuel cells can operate at higher efficiencies than combustion engines and can convert the chemical energy in the fuel directly to electrical energy with efficiencies capable of exceeding 60%.

Hydrogen-Oxygen Fuel Cell (Acidic Medium)

Electrons flow through an external circuit to produce electricity.

$H_2(g)$ is oxidised to form H^+



$O_2(g)$ is reduced and reacts with protons to form $H_2O(g)$ is then condensed to pure water

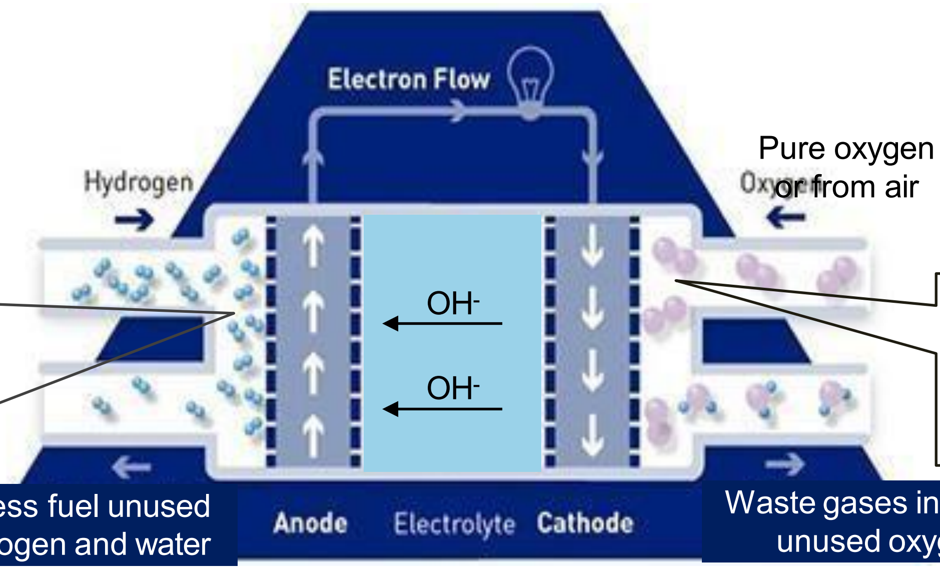
Anode:
$H_2(g) \rightarrow 2H^+(aq) + 2e^-$

Protons produced at the anode migrate through the Proton Exchange Membrane (PEM)

Cathode:
$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$

Hydrogen-Oxygen Fuel Cell (Basic Medium)

-H_2(g) is oxidised to form H^+ which immediately combine with OH^- to form H_2O.
-The electrons released flow through the external circuit from anode to cathode to produce electricity.



O_2(g) gain electrons and in combination with H_2O are reduced to OH^-

ANODE: $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$

CATHODE: $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$

OVERALL:
$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

CURRENT USAGE : ELECTRIC VEHICLES

Fuel Cell Electric Vehicles (FCEVs)

- FCEVs uses **fuel cell technology** to generate the power required to charge the battery and run the vehicle. The vehicle converts **chemical energy** into **electric energy**.

- FCEVs do not need to be plugged in like BEVs because the car itself produces the electricity needed for its functioning. Chemical energy is directly converted into electric energy in an FCEV.

- Some common examples of FCEVs are **Toyota Mirai, Hyundai Tucson FCEV, Honda Clarity Fuel Cell, Hyundai Nexo,** etc.

Exercise:

A fuel cell uses the chemical energy of fuels to cleanly and efficiently produce electricity. Each fuel cell has two electrodes; a negative anode and a positive cathode. By using hydrogen as the fuel and <u>an acid</u> as electrolyte, explain the reactions to produce electricity at these electrodes.

Answer:

Hydrogen enters the fuel cell through the anode and oxygen at cathode.

At the anode, hydrogen is oxidised to form H^+.

ANODE: $H_2(g) \rightarrow 2H^+(aq) + 2e^-$

The positively charged hydrogen pass through a porous membrane to the cathode.
The negatively charged electrons flow out of the cell and generate an electric current.

At the cathode, oxygen is reduced to water vapour then condensed to pure water.

CATHODE: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$

The overall equation :

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

Exercise:

A fuel cell uses the chemical energy of fuels to cleanly and efficiently produce electricity. Each fuel cell has two electrodes; a negative anode and a positive cathode.

By using hydrogen as the fuel and a basic electrolyte, explain the reactions to produce electricity at these electrodes.

Answer:

Hydrogen enters the fuel cell through the anode and oxygen at cathode.

At the anode, hydrogen is oxidised.
ANODE: $2H_2(g) + 4OH^{-}(aq) \rightarrow 4H_2O(l) + 4e^{-}$

The negatively charged electrons flow out of the cell and generate an electric current.

At the cathode, oxygen is reduced.
CATHODE: $O_2(g) + 2H_2O(l) + 4e^{-} \rightarrow 4OH^{-}(aq)$

The overall equation :
$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

Nernst Equation

- Used to calculate cell potential at non-standard conditions (E_{cell}).

$$E_{cell} = E^o_{cell} - \frac{RT}{nF}\ln Q \quad @ \quad E_{cell} = E^o_{cell} - \frac{2.303RT}{nF}\log Q$$

Substituting
$R = 8.314\ J\ K^{-1}\ mol^{-1}$
$F = 96500\ C mol^{-1}$
$T = 298.15\ K$

Thus, at 25°C

$$\boxed{\mathbf{E_{cell} = E^o_{cell} - \frac{0.0592}{n}\log Q}}$$

Cell potential at non-standard conditions

Standard cell potential

Moles of e^- transferred in the cell reaction

Reaction quotient

For the cell reaction,

$$aA(aq) + bB(aq) \longrightarrow cC(aq) + dD(aq)$$

$$E_{cell} = E^{o}_{cell} - \frac{0.0592}{n} \log \frac{[C]^{c}[D]^{d}}{[A]^{a}[B]^{b}} \longleftarrow \text{Reaction quotient, Q}$$

Note:
- ONLY those species exist in **aqueous** (concentration) and **gases** (pressure) are included in Q.
- Solids and liquids do not appear in Q expression.

Cell potential (E_{cell}) depends on the **concentrations** of ions in the cell, the **temperature** of solutions and **pressure** of gases involved.

Factors Affecting the Cell Potential, E_{cell}

The cell potential, E_{cell} is affected by:

1. **Concentration of the reactant and product**

$$\boxed{E_{cell} = E^{o}_{cell} - \frac{0.0592}{n} \log \frac{[C]^{c}[D]^{d}}{[A]^{a}[B]^{b}}}$$

- E_{cell} is inversely proportional to the [product] and directly proportional to the [reactant].

  i.e. Increase [reactant] or decrease [product] will increase the E_{cell} value.

2. **Temperature** $\boxed{E_{cell} = E^{o}_{cell} - \frac{RT}{nF} \ln Q}$

- Lowering or increasing the temperature will change the value of E_{cell}.

Factors Affecting the Cell Potential, E_{cell}

3. Type of half-cell

$E°_{cell} = E°_{cathode} - E°_{anode}$

- Using half-cell with more negative $E°_{red}$ value as anode will increase $E°_{cell}$ and E_{cell}.
- Using half-cell with more positive $E°_{red}$ value as cathode will increase the $E°_{cell}$ and E_{cell}.

Relationship Between Cell Potential, E_{cell} to Reaction Quotient, Q and Equilibrium Constant, K

$$_{\mathbf{cell}} - \frac{\mathbf{0.0592}}{\mathbf{n}}\mathbf{\log Q}$$

- At equilibrium, there is no net transfer of electrons.
- $E_{cell} = 0$
- Q becomes K (equilibrium constant)

Thus,

$$\boxed{E^{o}_{cell} = \frac{0.0592}{n}\log K}$$

The uses of Nernst Equation

- To determine cell potential (emf)
- To determine the concentration or partial pressure of a species in electrolytes
- To determine the equilibrium constant, K when $E_{cell} = 0$

Determine E_{cell} at non-standard conditions

Example:

Calculate the E_{cell} for the following cell:

$Cu(s) \mid Cu^{2+}(aq, 0.03M) \parallel Ag^{+}(aq, 0.002M) \mid Ag(s)$

Given: $E^{o}_{Cu^{2+}/Cu} = +0.34\ V \quad E^{o}_{Ag^{+}/Ag} = +0.80\ V$

Use **ABC rule** to identify anode and cathode

Solution:

Step 1: Write the half-cell equations and the overall equation

Anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$ $\boxed{n=2}$

Cathode: $[Ag^{+}(aq) + e^{-} \rightarrow Ag(s)] \times 2$

Overall: $Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

Step 2 : Calculate E°_{cell}

$$E^{o}_{cell} = E^{o}_{cathode} - E^{o}_{anode}$$
$$= +0.80 - (+0.34)$$
$$= +0.46\ V$$

Step 3 :
Calculate the E_{cell} using Nernst equation

$$E_{cell} = E^{o}_{cell} - \frac{0.0592}{n}\log\frac{[Cu^{2+}]}{[Ag^{+}]^{2}}$$

$$= +0.46 - \frac{0.0592}{2}\log\frac{(0.03)}{(0.002)^{2}}$$

$$= +\ 0.345\ V$$

Determine the concentration or partial pressure of a species in electrolytes

Example:

The E_{cell} for the following electrochemical cell is +0.34 V at 25^{o}C.

$$Zn(s)\ |\ Zn^{2+}(aq,\ 0.1\ M)\ ||\ Ni^{2+}(aq,\ x\ M)\ |\ Ni(s)$$

Determine the value of *x*.

Use **ABC rule** to identify anode and cathode

Given: $E^{o}_{Zn^{2+}/Zn} = -\ 0.76\ V$, $E^{o}_{Ni^{2+}/Ni} = -\ 0.25\ V$

Solution:

Step 1:

Anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode: $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$

$\boxed{n=2}$

Overall: $Zn(s) + Ni^{2+}(aq) \rightarrow Zn^{2+}(aq) + Ni(s)$

Step 2:

$$E^o_{cell} = E^o_{cathode} - E^o_{anode}$$
$$= -0.25 - (-0.76)$$
$$= +0.51\ V$$

Step 3:

$$E_{cell} = E^o_{cell} - \frac{0.0592}{n} \log \frac{[Zn^{2+}]}{[Ni^{2+}]}$$

$$+0.34 = +0.51 - \frac{0.0592}{2} \log \frac{(0.1)}{[Ni^{2+}]}$$

$[Ni^{2+}] = 1.81 \times 10^{-7}$ M = **x**

Determine the equilibrium constant, K

Example: Determine the equilibrium constant (*K*) for the following reaction.

$$Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$$

Given: $E^{o}_{Cu^{2+}/Cu} = +0.34\ V$, $E^{o}_{Ag^{+}/Ag} = +0.80\ V$

Compare the changes in oxidation number to identify anode and cathode

Solution:

Step 1: Write the half-cell equations and the overall equation

Anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$

Cathode: $[Ag^{+}(aq) + e^{-} \rightarrow Ag(s)] \times 2$

$\boxed{n=2}$

Overall: $Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

Solution:

Step 2:
Calculate $E°_{cell}$

$$E^o_{cell} = E^o_{cathode} - E^o_{anode}$$
$$= +0.80 - (+0.34)$$
$$= +0.46\ V$$

Step 3:
Calculate K
using Nernst
equation

$$E_{cell} = E^o_{cell} - \frac{0.0592}{n}\log Q$$

At equilibrium, $Q = K$ and $E_{cell} = 0$

$$E^o_{cell} = \frac{0.0592}{n}\log K$$

$$+0.46 = \frac{0.0592}{2}\log K \qquad \mathbf{K = 3.472 \times 10^{15}}$$

Predict The Spontaneity of a Cell

$E_{cell} > 0$ (+ve)	The reaction is spontaneous
$E_{cell} < 0$ (-ve)	The reaction is not spontaneous
$E_{cell} = 0$	The reaction is at equilibrium

Example: Predict the spontaneity of this galvanic cell.

$Ni(s) \mid Ni^{2+}(aq, 0.02M) \parallel Cl^{-}(aq, 0.1M) \mid Cl_2(g, 1atm) \mid Pt(s)$

Given: $E^o_{Ni^{2+}/Ni} = -\ 0.25\ V$

$E^o_{Cl_2/Cl^-} = +\ 1.36\ V$

Use **ABC rule** to identify anode and cathode

Solution:

Step 1:

Anode: $Ni(s) \rightarrow Ni^{2+}(aq) + 2e^-$

n=2

Cathode: $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$

Overall: $Ni(s) + Cl_2(g) \rightarrow Ni^{2+}(aq) + 2Cl^-(aq)$

Step 2:

$$E^o_{cell} = E^o_{cathode} - E^o_{anode}$$

$$= +1.36 - (-0.25)$$

$$= +1.61\ V$$

Step 3:

$$E_{cell} = E^o_{cell} - \frac{0.0592}{n}\log\frac{[Ni^{2+}][Cl^-]^2}{P_{Cl_2}}$$

$$= +1.61 - \frac{0.0592}{2}\log\frac{(0.02)(0.1)^2}{1} = +1.72\ V$$

Since E_{cell} **is positive**, **the reaction is spontaneous**.

3.2 ELECTROLYTIC CELL

Electrolysis

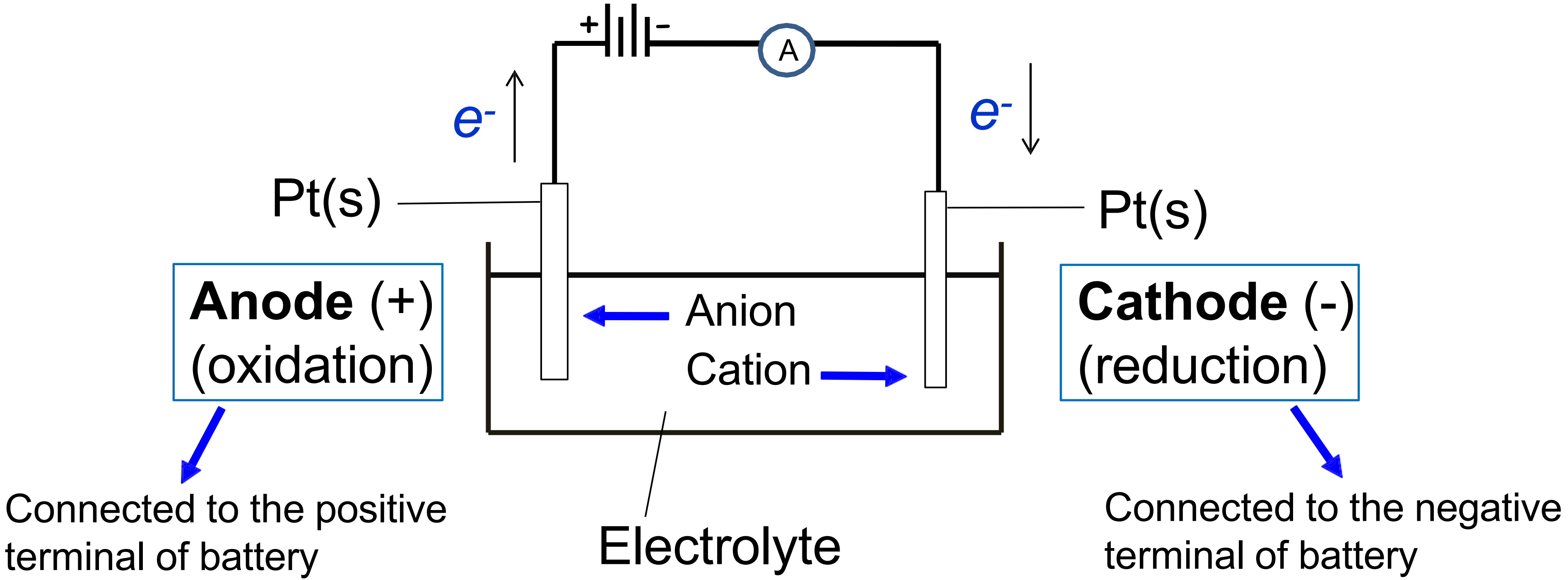
- Is a chemical process that uses electricity to cause a **non-spontaneous** redox reaction to occur.

Electrolytic Cell

- It is made up of 2 electrodes immersed in an electrolyte.
- A direct current is passed through the electrolyte from an external source.
- Molten salt and aqueous ionic solution are commonly used as electrolytes.

Electrolytic Cell

- Anions are attracted to anode (positive electrode).
- Cations are attracted to cathode (negative electrode).



Electrolytic cell vs Galvanic cell

Electrolytic Cell	**Galvanic Cell**
The oxidation and reduction are non-spontaneous.	The oxidation and reduction are spontaneous.
Electrical energy is converted to chemical energy.	Chemical energy is converted to electrical energy.
The anode is positive electrode and the cathode is negative electrode.	The anode is negative electrode and the cathode is positive electrode.

Similarities between Electrolytic cell and Galvanic cell

- **Oxidation** occurs at **anode** and **reduction** occurs at **cathode**.
- **Anions** move **to anode** and **cations** move **to cathode**.
- **Electrons flow** from **anode to cathode** in the external circuit.

Electrolysis of Molten Salt Using Inert Electrodes

- Molten salts contains only cations and anions.
- In the electrolytic cell:
  - **anion** attracted to the **anode** (positive electrode) and undergoes **oxidation**.
  - **cation** attracted to the **cathode** (negative electrode) and undergoes **reduction**.
- Electrolysis of molten salt requires high temperature.

Example: Electrolysis molten NaCl

Ions present: Na^+ and Cl^-

Anode: $2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$

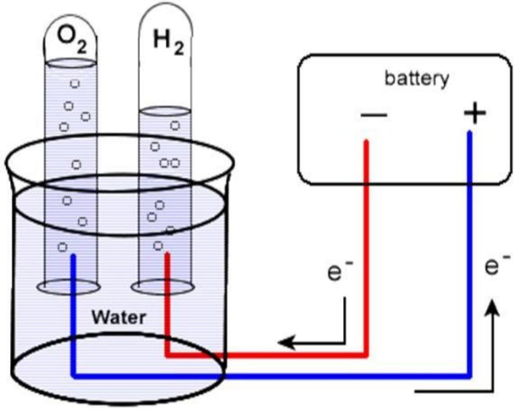
Cathode: $[Na^+(l) + e^- \rightarrow Na(s)] \times 2$

Overall: $2Na^+(l) + 2Cl^-(l) \rightarrow 2Na(s) + Cl_2(g)$

- At anode, **chlorine gas (Cl_2)** is liberated.
- At cathode, **sodium metal (Na)** is deposited.

Electrolysis of Water Using Inert Electrodes

- Decomposition of water, H_2O to O_2 and H_2.



<u>Anode</u> (+):

Oxidation of water molecules (release e^-)

$$\mathbf{2H_2O(l) \rightarrow O_2(g) \; + \; 4H^+(aq) \; + \; 4e^-}$$

<u>Cathode</u> (-):

Reduction of water molecules (accept e^-)

$$\mathbf{2H_2O(l) + \; 2e^- \rightarrow H_2(g) + \; 2OH^-(aq)}$$

Overall: $\mathbf{2H_2O(l) \rightarrow O_2(g) \; + \; 2H_2(g)}$

- At anode, **oxygen gas (O_2)** is liberated.
- At cathode, **hydrogen gas (H_2)** is liberated.

Electrolysis of an Aqueous Salt Solution

- Aqueous salt solution contains water molecule, anion and cation.
- Water is an electro-active substance that may be oxidised or reduced.
- At anode, water competes with anions to undergo oxidation.
- At cathode, water competes with cations to undergo reduction.
- Hence, the product formed at each electrodes depend on which species are selected to be discharged.

Factors on the selective discharge of a species at the electrode:

Three Factors

Standard Reduction Potential, $E°_{red}$ of the species

- Anions species with **more negative $E°_{red}$ value** will be discharged first **at anode**.
- Cations with **more positive $E°_{red}$ value** will be discharged first **at cathode**.

Concentration of the species

- Species with **higher concentration** will be discharged first if the **$E°_{red}$ values** of two competing species are **very close**.
- If the difference between the **$E°_{red}$ values** of two competing species are **too large**, the concentration factor is **no longer important.**

Nature of electrodes

- Inert electrode: such as C or Pt.
- Active electrode: such as Cu or Ag.

The following steps may help in predicting the selective discharge of a species at the electrode:

Step 1: Identify the movement of ions present.

Step 2: At the anode:

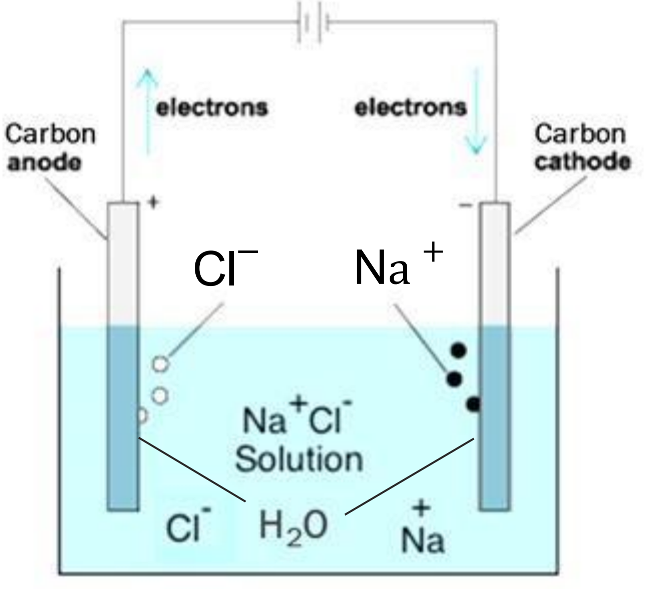
- For an inert electrode: Choose an ion with **lower (more negative) E°_{red}** value to undergo oxidation.
- For the concentrated electrolyte, ion with **higher concentration** will be discharged first if the **E°_{red} values** of two competing species are **very close**.
- For an active electrode: the electrode dissolves to form metallic ion (oxidation).

Step 3: At the cathode:

- For an inert electrode: Choose an ion with a **higher (more positive) E^o_{red}** value to undergo reduction.
- For the concentrated electrolyte, ion with **higher concentration** will be discharged first if the **E°_{red} values** of two competing species are **very close**.

Electrolysis of a Aqueous NaCl Solution Using Inert Electrodes

- The aqueous solution of NaCl contains water molecules, Na^+ and Cl^- ions.
- Na^+ ion and H_2O molecules are attracted to the cathode.
- Cl^- ion and H_2O molecules are attracted to the anode.
- The product of electrolysis of aqueous NaCl depends on the **concentration of the electrolyte** and **$E°_{red}$ of species** involved.



Electrolysis of a dilute aqueous NaCl solution

Anode (+)

✓ E^o_{red} for H_2O is more negative.
✓ H_2O is selected for oxidation.

Ions attracted to anode : $Cl^- \,\&\, H_2O$

$E^o_{red} = +1.36V$: $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$

$E^o_{red} = +1.23V$: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$

Cathode (–)

✓ E^o_{red} for H_2O is more positive.
✓ H_2O is selected for reduction.

Ions attracted to cathode : $Na^+ \,\&\, H_2O$

$E^o = -2.71V$: $Na^+(aq) + e^- \rightarrow Na(s)$

$E^o = -0.83V$: $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$

$$\text{Anode: } 2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$$

$$\text{Cathode: } [2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)] \times 2$$

$$6H_2O(l) \rightarrow O_2(g) + 2H_2(g) + \underbrace{4H^+(aq) + 4OH^-(aq)}_{4H_2O(l)}$$

$$\text{Overall equation: } 2H_2O(l) \rightarrow O_2(g) + 2H_2(g)$$

- At anode, **oxygen gas (O_2)** is liberated.
- At cathode, **hydrogen gas (H_2)** is liberated.

Electrolysis of a concentrated aqueous NaCl solution

✓ In concentrated solution, chloride ions is selected for oxidation because of its high concentration.

Anode (+)

Ions attracted to anode : Cl^{-} & H_2O

$E^o_{red} = +1.36V$: $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$

$E^o_{red} = +1.23V$: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$

- ✓ The E°_{red} values of the two competing species are **very close**.
- ✓ The **concentration** factor is **significant**.

Cathode (–)

Ions attracted to cathode : Na^+ & H_2O

✓ E^o_{red} for H_2O is more positive.
✓ H_2O is easier to be reduced.

$E^o = -2.71V$: $Na^+(aq) + e^- \rightarrow Na(s)$

$E^o = -0.83V$: $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$

- ✓ The difference between the E°_{red} values of the two competing species is **too large**.
- ✓ The **concentration** factor is **no longer important**.

$$\text{Anode: } 2Cl^-(aq) \rightarrow Cl_2\,(g) \;+\; 2e^-$$

$$\text{Cathode: } 2H_2O(l) + 2e^- \rightarrow \; H_2(g) + 2OH^-(aq)$$

$$\text{Overall reaction: } 2H_2O(l) + 2Cl^-(aq) \;\rightarrow Cl_2(g) + H_2(g) + 2OH^-(aq)$$

- At anode, **chlorine gas (Cl_2)** is liberated.
- At cathode, **hydrogen gas (H_2)** is liberated.

Electrolysis of an aqueous Na_2SO_4 solution

- Na_2SO_4 aqueous solution contains Na^+ , SO_4^{2-} and H_2O molecules.
- During electrolysis:
  - $SO_{4_{2-}}$ ion and H_2O are attracted to the anode
  - Na^+ ion and H_2O are attracted to the cathode

✓ E^o_{red} for H_2O is more negative.
✓ H_2O is selected for oxidation.

Anode (+)

Ions attracted to anode : SO_4^{2-} & H_2O

$E^o_{red} = +2.01$ V : $S_2O_8^{2-}(aq) + 2e^- \rightarrow 2SO_4^{2-}(aq)$

$E^o_{red} = +1.23$V : $O_2(g) + 4H^+(aq) + 4e \rightarrow 2H_2O(l)$

✓ E^o_{red} for H_2O is more positive.
✓ H_2O is selected for reduction.

Cathode (–)

Ions attracted to cathode : Na^+ & H_2O

$E^o = -2.71$V : $Na^+(aq) + e \rightarrow Na(s)$

$E^o = -0.83$V : $2H_2O(l) + 2e \rightarrow H_2(g) + 2OH^-(aq)$

Anode: $2H_2O\ (l) \rightarrow O_2\ (g) + 4H^+\ (aq) + 4e^-$

Cathode: $[2H_2O\ (l) + 2e^- \rightarrow H_2\ (g) + 2OH^-\ (aq)\ ] \times 2$

Overall equation: $2H_2O\ (l) \rightarrow\ O_2(g) + 2H_2(g)$

Anode: **O_2 gas** is liberated and solution become acidic because H^+ ions are formed.

Cathode: **H_2 gas** is liberated and solution become basic because OH^- ions are formed.

Factors affecting the products of electrolysis: Nature of electrodes

- Inert electrodes (e.g. graphite, platinum) are not involved in the reaction.

- Active electrodes (anode) such as copper or silver will dissolve to form metallic ions if the electrolyte contains the ions of the same metal.

Example:

Electrolysis of $CuSO_4(aq)$ using Cu electrode

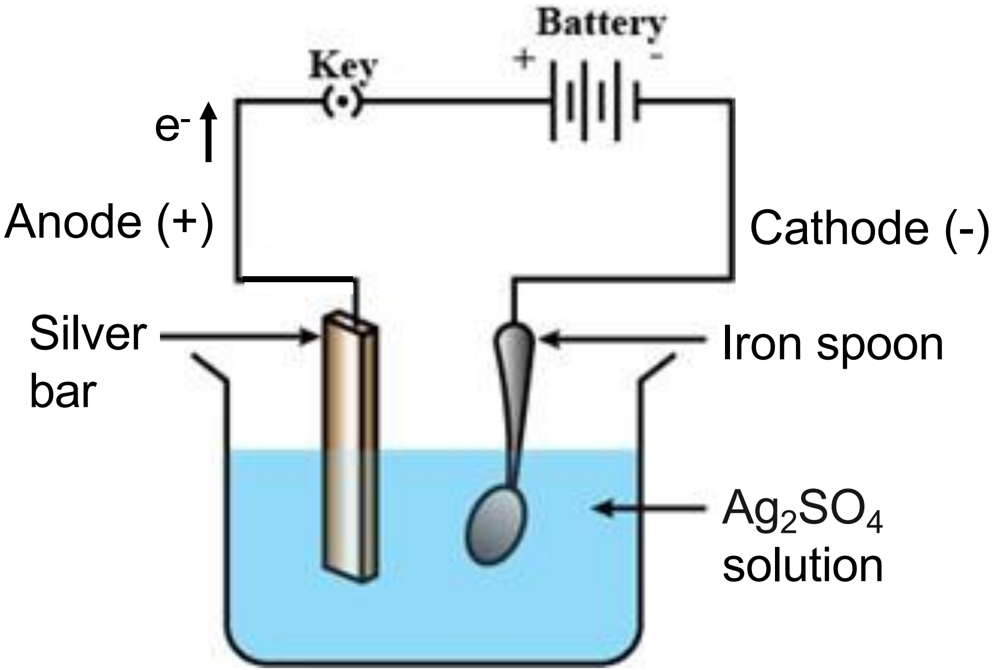
Anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

Cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Overall : $Cu^{2+}(aq) + Cu(s) \rightarrow Cu(s) + Cu^{2+}(aq)$

Uses of Electrolysis: Electroplating

Example : Silver electroplating on an iron, Fe spoon.



<u>Anode</u>

$Ag(s) \rightarrow Ag^{+}(aq) + e^{-}$

- The silver bar (anode) **dissolves** or becomes thin.

<u>Cathode</u>

$Ag^{+}(aq) + e^{-} \rightarrow Ag(s)$

- The iron spoon (cathode) is **coated** with a thin layer of silver.

- At the **anode**, electrons are removed from the **Ag** metal, **oxidising** it to the **Ag^+**.
- Ag metal dissolves as Ag^+ into the solution, supplying replacement Ag^+ for that which has been plated out.
- Fe spoon (**cathode**) attracts the positively charged Ag^+.
- When the Ag^+ reaches the Fe spoon, electrons flow from Fe to **reduce** the **Ag^+ into metallic form**.
- Metallic coating is formed around/on the Fe spoon.

Faraday's Law

- Describes the relationship between the amount of electricity passed through an electrolytic cell and the amount of substances produced at electrode.

Faraday's First Law of Electrolysis

- States that the amount of a substance produced or consumed during electrolysis is directly proportional to the amount of electricity that passes through the electrical circuit of the cell.

$$m \propto Q$$

m = mass of substance discharged

Q = electric charge in coulombs (C)

Electrical Charge, Q

$$\mathbf{Q = It}$$

Q = Electric charge in coulombs, C
I = Current in ampere, A
T = Time, s

Faraday's Constant, F

1 Faraday is the amount of electric charge carried by one mol of e^-.

1 mol $e^- \equiv$ 1 F $\equiv$ 96 500 C

Example:

An aqueous solution of $CuSO_4$ is electrolysed using a current of 0.150 A for 5 hours. Calculate the mass of copper deposited at the cathode.

Solution:

Step 1: Calculate Q

$$\text{Electric charge, } Q = I \times t$$

$$= (0.150\ \text{A}) \times (5 \times 60 \times 60)\text{s}$$

$$= 2\ 700\ \text{C}$$

Step 2: Write half-equation at cathode

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

2 mol of electron ≡ 2F ≡ 96 500 C × 2 = 193 000 C ≡ 1 mol of Cu

Step 3: Calculate the number of mole of Cu produced

$$193\,000\ \text{C} \equiv 1\ \text{mol Cu}$$

$$\therefore\ 2\,700\ \text{C} \equiv \frac{2\,700 \times 1}{193\,000}$$

$$= 0.0140\ \text{mol Cu}$$

Step 4: Calculate mass of Cu

Mass of copper deposited $= 0.0140 \text{ mol} \times 63.6 \text{ g mol}^{-1}$

$= 0.89$ g

THE END